INCH-POUND

MIL-PRF-1/1174E
29 June 1999
SUPERSEDING
MIL-E-1/1174D
17 June 1975

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 6303

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Rectifier or shunt diode, high-vacuum.

Outline: See figure 1.

Mounting position: Vertical, base down.

Weight: 10 ounces (283.5 grams) nominal.

ABSOLUTE RATINGS:

Parameter: Unit:	Ef V ac	epx kv	ib a	Ib mA dc	Ieff A ac	Pp W
Rectifier						
Maximum:	11.8	40	2.5	700	---	550
Minimum:	11.2	---	---	---	---	---
Shunt diode						
Maximum:	12.5	33	50	---	1.25	500
Minimum:	11.9	---	---	---	---	---
Test conditions:	11.5	---	---	---	---	---

See footnotes at end of table I.

GENERAL:

Qualification - Required.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:15Z
Check the source to verify that this is the current version before use.

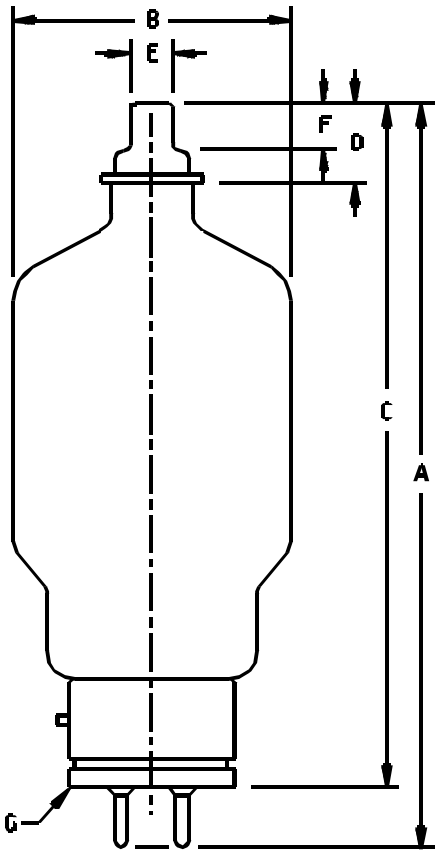
TABLE I. Testing and inspection.

Inspection	Method	Notes	Conditions	Acceptance level	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
Conformance inspection, part 1									
Filament current	1301	---		0.65	II	If	14.00	16.25	A ac
Electrode current (anode)	1256	---	Eb = 500 V dc	0.65	II	Ib	750	---	mA dc
Peak emission	1231	---	eb = 8.0 kv; Ef = 12.2 V ac	0.65	II	is	25	---	a
Operation of rectifiers (1)	1353	1/	epx = 40 kv; Ib = 700 mA dc; ib = 2.5 a	0.65	II	---	---	---	---
Conformance inspection, part 2									
Low-frequency vibration	1031	---	No voltages applied	---	---	---	---	---	---
Shock	1041	---	Hammer angle = 20°	---	---	---	---	---	---
Operation of rectifiers (2)	1353	1/ 2/	Ef = 12.2 V ac	---	---	---	---	---	---
Conformance inspection, part 3									
Life test	---	---	Group C; operation of rectifiers (1); t = 500 hours	---	---	---	---	---	---
Life-test end point:	---								
Peak emission	1231	---		---	---	is	20	---	a

1/ The tube shall operate for 2 minutes without arcing or sign of gas during a test interval not to exceed 5 minutes.

2/ The tube shall be operated as a shunt diode in a thyratron modulator. The circuit constants shall be chosen to give the following diode operating conditions: epx = 33 kv; ib = 50 a; and effective current (Ieff) = 1.25 A ac.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:15Z
Check the source to verify that this is the current version before use.



Pin connections	
Pin No.	Element
1	nc
2	f
3	f
4	nc
Cap	a

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	9.250	9.750	234.95	247.65
B	---	3.625	---	92.08
C	8.472	9.032	215.19	229.41
Conformance inspection, part 3 (see note)				
D	.969	1.031	24.62	26.19
E	.559	.573	14.20	14.55
F	.594	.656	15.09	16.66
G	Base: A4-18 (EIA)			

NOTE: Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of conformance inspection, part 2, code level D, with an acceptance level of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 6303.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:15Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - MI
Navy - AS, CG, MC, OS
Air Force - 17, 19, 99

Preparing activity:
DLA - CC

(Project 5960-3530)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:15Z
Check the source to verify that this is the current version before use.